

RESEARCHES ON THE PRIME DIVISORS OF A CLASS OF FORMULAE OF THE FOURTH DEGREE.

1st Memoir.

In the literary announcements of Göttingen (11 April 1825) one finds the extract of a memoir of indeterminate analysis which Mr. GAUSS presented to the Royal Society of Sciences of that city, but which has not yet been printed. This memoir is the first of a series of memoirs which the illustrious author of the *Disquisitiones arithmeticae* proposes to give on the theory of biquadratic residues, and its object is to determine the distinctive characters of the prime divisors of the formula $x^4 - 2$. The author establishes in it two extremely elegant theorems which can serve to decide whether a prime number, divisor of $x^2 - 2$, divides or does not divide the preceding formula. Having learned, in the course of the year just ended, of the cited extract, which contains only the statements of the two theorems just mentioned and of some auxiliary propositions, I wished to demonstrate on my side the beautiful theorems discovered by Mr. GAUSS. The researches which I made with this aim led me to find a proof founded on extremely simple considerations and probably entirely different from that of Mr. GAUSS, which appears to require very delicate and rather extensive preliminary researches. I then applied analogous considerations to other questions and particularly to the search for the properties which distinguish the prime divisors of the formula $\alpha x^4 + \beta x^2 + \gamma$; and I thus arrived at a large number of interesting theorems. The rapid exposition of part of the results to which these researches have led me is the object of the present memoir. I begin by setting down some definitions and by stating some theorems very easy to establish and on which we shall have to rely in what follows.

1.

“If one can assign to the indeterminate x a value such that $x^4 - A$ becomes divisible by B , A will be called a biquadratic residue with respect to B .”

It is easy to see that, if a number A is a biquadratic residue with respect to a number B , or in other words, if B is a divisor of $x^4 - A$, each prime factor of B will likewise be a divisor of $x^4 - A$, and reciprocally, that if this condition holds with respect to every prime factor of the number B , B will itself be a divisor of $x^4 - A$. Thus, when it is a question of assigning all the numbers which divide the formula $x^4 - A$, one can confine oneself to considering prime numbers.

It is no less evident that, for a number to divide the formula $x^4 - A$, it is necessary that this number be a divisor of $x^2 - A$. There are therefore only the prime divisors of this latter formula to examine, that is to say, the prime numbers with respect to which A is a quadratic residue. One may add that, relative to those among these latter which are of the form $4n + 3$, the question presents no difficulty; for one verifies by a very simple reasoning that every prime number $4n + 3$, divisor of $x^2 - A$, also divides the formula $x^4 - A$.

Let p be a prime number $4n + 1$, divisor of $x^2 - A$, A designating a positive or negative number not divisible by p ; one has, as is known, $A^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. From this one concludes $A^{\frac{p-1}{4}} \equiv \pm 1 \pmod{p}$ and one proves easily that the upper or lower sign will hold according as p divides or does not divide the formula $x^4 - A$. Thus one may state this theorem:

“ A designating a number which is a quadratic residue with respect to the prime number $p = 4n + 1$, one will have $A^{\frac{p-1}{4}} \equiv 1$ or $A^{\frac{p-1}{4}} \equiv -1 \pmod{p}$. In the first case, A will be a biquadratic residue with respect to p , in the second, A will be a biquadratic non-residue with respect to p .”

If this theorem is applied to the case where $A = -1$, one will find that -1 is a biquadratic residue with respect to the prime numbers $8n + 1$, and on the contrary a non-residue relative to those of the form $8n + 5$. From the preceding theorem one easily deduces the following other theorem, in whose statement one supposes, as in all that follows, that p is a prime number $4n + 1$, and that A and A' designate numbers not divisible by p and which are both quadratic residues with respect to p .

“If the numbers A and A' are both biquadratic residues or both biquadratic non-residues with respect to p , the product AA' will be a biquadratic residue with respect to p ; if, on the contrary, one of the numbers A and A' is a residue and the other a biquadratic non-residue relative to p , AA' will be a biquadratic non-residue with respect to p .”

Taking $A' = -1$, and having regard to what precedes, one will see that, if p is of the form $8n + 1$, A will at the same time as $-A$ be a biquadratic residue or non-residue, and that, on the contrary, if p is of the form $8n + 5$, one of the numbers A and $-A$ will be a residue and the other a biquadratic non-residue with respect to p .

2.

After establishing these preliminaries, we shall deal with the search for the characters which distinguish the prime divisors of the formula $x^4 - 2$. It is known that the prime divisors of $x^2 - 2$ are of one of these two forms: $8n + 1$, $8n + 7$, and that reciprocally every prime number of one of these forms divides the formula $x^2 - 2$. According to what we said in the preceding paragraph, we need not take account of the latter of these two forms, and it will suffice to consider the prime numbers $8n + 1$. Let p be a prime number of this kind, and put, as one is permitted to do, $p = t^2 + 2u^2$, where u will be even and t odd. Let $u = 2^\nu k k' k'' \dots$, 2^ν being the highest power of 2 which divides u , and $k, k', k'', \dots$ designating odd prime numbers, several of which may be equal to one another. The equation $t^2 + 2u^2 = p$ immediately gives $t^2 \equiv p \pmod{k}$. The number p is therefore a quadratic residue with respect to k , which we shall write thus: $\left(\frac{p}{k}\right) = 1$, adopting the notation used by Mr. LEGENDRE. Recall that, if c designates a prime number and M any number not divisible by c , this illustrious geometer uses the sign $\left(\frac{M}{c}\right)$ to designate the remainder which one obtains by dividing by c the power $M^{\frac{c-1}{2}}$, a remainder which is known to be equal to 1 or -1 , according as M is or is not a quadratic residue with respect to c . Applying to the relation $\left(\frac{p}{k}\right) = 1$ the theorem known under the name law of reciprocity (the *theorema fundamentale* of Mr. GAUSS), one will have, p being

of the form $4n + 1$, $\left(\frac{k}{p}\right) = 1$. One finds in the same way:

$$\left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

On the other hand, since p is of the form $8n + 1$, one also has $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$. Multiplying this last relation by all the preceding ones, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = 1.$$

Now consider the simple factors of the odd number t , which we shall divide into two classes. The first class will comprise the prime divisors of one of these two forms: $8n + 1$, $8n + 7$, and the numbers belonging to it will be designated by $g, g', g'', \dots$; the second class will be composed of numbers $h, h', h'', \dots$, contained in these two forms: $8n + 3$, $8n + 5$. One first has:

$$t = gg'g'' \dots \times hh'h'' \dots,$$

and one will then conclude from the equation $p = t^2 + 2u^2$:

$$\begin{aligned} \left(\frac{2p}{g}\right) = 1, \quad \left(\frac{2p}{g'}\right) = 1, \quad \left(\frac{2p}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{2p}{h}\right) = 1, \quad \left(\frac{2p}{h'}\right) = 1, \quad \left(\frac{2p}{h''}\right) = 1, \quad \text{etc.} \end{aligned}$$

On the other hand, by known theorems one has:

$$\begin{aligned} \left(\frac{2}{g}\right) = 1, \quad \left(\frac{2}{g'}\right) = 1, \quad \left(\frac{2}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{2}{h}\right) = -1, \quad \left(\frac{2}{h'}\right) = -1, \quad \left(\frac{2}{h''}\right) = -1, \quad \text{etc.} \end{aligned}$$

If one now compares these relations with the preceding ones, one will find:

$$\begin{aligned} \left(\frac{p}{g}\right) = 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{p}{h}\right) = -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.} \end{aligned}$$

The application of the law of reciprocity to these last relations gives:

$$\begin{aligned} \left(\frac{g}{p}\right) = 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.} \\ \left(\frac{h}{p}\right) = -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.} \end{aligned}$$

hence, on multiplying:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p} \right) = \left(\frac{t}{p} \right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number. But it is easy to see, from the equation $t = gg'g'' \dots \times hh'h'' \dots$, that the first case will occur when t is of one of these forms: $8n + 1, 8n + 7$, and the second when t is contained in one of these: $8n + 3, 8n + 5$. Thus one has:

$$\begin{aligned} \left(\frac{t}{p} \right) &= 1, & \text{when } t &= 8n + 1 \quad \text{or} \quad 8n + 7, \\ \left(\frac{t}{p} \right) &= -1, & \text{when } t &= 8n + 3 \quad \text{or} \quad 8n + 5. \end{aligned}$$

Return to the equation $t^2 + 2u^2 = p$ and put it in the form of a congruence:

$$t^2 \equiv -2u^2 \pmod{p}.$$

Raising both sides to the power $\frac{p-1}{4}$, one will have $\left(\frac{p-1}{4} \right.$ being even):

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

or, what comes to the same thing, having proved that $\left(\frac{u}{p} \right) = 1$:

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} \pmod{p}.$$

Thus one sees that ± 2 (the double sign may be put, since $p = 8n + 1$) is or is not a biquadratic residue with respect to p according as one has $\left(\frac{t}{p} \right) = 1$ or $\left(\frac{t}{p} \right) = -1$. Comparing this result with what precedes, one has the theorem:

“ p designating a prime number $8n + 1$, if one writes $p = t^2 + 2u^2$, I say that ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of these forms: $8n + 1, 8n + 7$ or of one of these: $8n + 3, 8n + 5$.”

This is the first of the two theorems of Mr. GAUSS mentioned in the preamble of this memoir. The second of these theorems is relative to the decomposition of the number p into two squares, and can easily be deduced from the one just established.

Let $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4) and equate this value of p with the one we have just considered.

We shall thus have:

$$p = t^2 + 2u^2 = \varphi^2 + \psi^2,$$

and, after transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 - 2u^2.$$

Since φ is odd, the greatest common divisor of φ and u will be odd; designate it by m and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the preceding equation changes it into:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 - 2u'^2).$$

The odd number $t + \psi$ is thus composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 - 2u'^2$; and the same is true of $t - \psi$, whose factors we denote by F and L , L being allowed to be negative. We therefore have the equations:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 - 2u'^2 &= KL. \end{aligned}$$

It is easy to verify that the numbers E and F are relatively prime. Indeed, let δ be a prime divisor of E , and suppose that δ also divides F . The number δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two numbers. On the other hand, from the fact that δ is a prime divisor of E , it follows successively, having regard to the equations $EF = m^2$, $u = mu'$, that m^2 , m , and u are multiples of δ . The numbers t and u would therefore have the common divisor δ , and $p = t^2 + 2u^2$ would not be a prime number.

The product of the numbers E and F , which are relatively prime, being a square, each of them is also a square. Let $E = e^2$; then $t + \psi = e^2K$. The odd square e^2 is of the form $8n + 1$, and K , as an odd divisor of $\varphi'^2 + 2u'^2$ (where φ' and u' are relatively prime), is of one of these forms: $8n + 1$, $8n + 7$. Thus the number $t + \psi$ will itself be of one of the forms $8n + 1$, $8n + 7$. The number ψ , which we know to be divisible by 4, will be of the form $8n$ or of the form $8n + 4$. It follows from the preceding that, in the first case, t will be of one of these two forms: $8n + 1$, $8n + 7$, and in the second of one of these: $8n + 3$, $8n + 5$. Comparing this result with the preceding theorem, one has this other theorem:

“ p designating a prime number $8n + 1$, and having written $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4), ± 2 will or will not be a biquadratic residue with respect to p according as ψ is of the form $8n$ or of the form $8n + 4$.”

There is a third theorem suitable for deciding whether ± 2 is or is not a biquadratic residue relative to a prime number $p = 8n + 1$, which may be stated as follows:

“Having made, in any manner, $p = t^2 - 2u^2$, ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of the forms $8n + 1$, $8n + 3$, or of one of these: $8n + 5$, $8n + 7$.”

We shall not stop to prove this theorem, which can be established directly and by considerations analogous to those on which the proof of the first of the two preceding theorems is founded. One can also deduce it from each of the preceding theorems almost as one has just passed from the first to the second.

3.

We now pass to more general considerations. Let b be a prime number $4n + 3$, p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$, and let us propose to decide whether $-b$ is or is not a biquadratic residue with respect to p . It is easy to see that, if p can be put under the form $t^2 - bu^2$, one can always do so in such a way that u is even, and consequently t odd^{*)}. Let therefore $p = t^2 - bu^2$, and put $u = 2^\nu k k' k'' \dots$, where $k, k', k'', \dots$ are the simple odd factors of u . One concludes immediately from the preceding equation:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \text{etc.}$$

The application of the reciprocity law then gives, p being of the form $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

Multiplying these relations together and with the identical relation $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$,

^{*)}To prove that this is always possible, we shall show that, if one has $t^2 - bu^2 = p$, t being even, it is easy to deduce from the values of t and u other numbers t' (odd) and u' which likewise satisfy the equation $t'^2 - bu'^2 = p$. Let r and s be the least numbers such that $r^2 - bs^2 = 1$; I say that r will be even. Indeed, it is known that, if b designates a prime number $4n + 3$, the equation $\rho^2 - b\sigma^2 = \pm 2$ is always possible, and that, ρ and σ being supposed the least numbers which satisfy it, one has $r = b\sigma^2 \pm 1$, $s = \rho\sigma$ (*Theory of Numbers*, no. 44.45). From this, and from the fact that the numbers ρ and σ are evidently both odd, it follows that r is an even number. This being so, if one multiplies together the equations $r^2 - bs^2 = 1$, $t^2 - bu^2 = p$, one obtains $(rt \pm bsu)^2 - b(ru \pm st)^2 = p$, and one will easily see that $rt \pm bsu$ is an odd number.

one has the result:

$$(\alpha) \quad \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

Now decompose the odd number t into its simple factors and divide these factors into two classes. Those of the first class will be designated by $g, g', g'', \dots$, and are such that:

$$(\beta) \quad \left(\frac{-b}{g}\right) = 1, \quad \left(\frac{-b}{g'}\right) = 1, \quad \left(\frac{-b}{g''}\right) = 1, \quad \text{etc.}$$

As for those which form the second class and which we shall designate by $h, h', h'', \dots$, they are such that:

$$(\beta') \quad \left(\frac{-b}{h}\right) = -1, \quad \left(\frac{-b}{h'}\right) = -1, \quad \left(\frac{-b}{h''}\right) = -1, \quad \text{etc.}$$

The product of all these numbers is equal to t , that is:

$$t = gg'g'' \dots \times hh'h'' \dots$$

Inspection of the equation $t^2 - bu^2 = p$ gives the results:

$$\begin{aligned} \left(\frac{-bp}{g}\right) &= 1, \quad \left(\frac{-bp}{g'}\right) = 1, \quad \left(\frac{-bp}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{-bp}{h}\right) &= 1, \quad \left(\frac{-bp}{h'}\right) = 1, \quad \left(\frac{-bp}{h''}\right) = 1, \quad \text{etc.} \end{aligned}$$

Comparison of these relations with the preceding ones then gives:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.} \end{aligned}$$

Applying now the reciprocity law, one obtains:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.} \end{aligned}$$

Multiplying these relations together, one will have:

$$\left(\frac{gg'g'' \dots \times hh'h''}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign holding according as the numbers $h, h', h'', \dots$ are even or odd in number. On the other hand, if one applies the reciprocity law to the relations (β) and (β') , one obtains, b being of the form $4n + 3$:

$$\begin{aligned} \left(\frac{g}{b}\right) &= 1, & \left(\frac{g'}{b}\right) &= 1, & \left(\frac{g''}{b}\right) &= 1, & \text{etc.} \\ \left(\frac{h}{b}\right) &= -1, & \left(\frac{h'}{b}\right) &= -1, & \left(\frac{h''}{b}\right) &= -1, & \text{etc.} \end{aligned}$$

equalities which, when multiplied together, give:

$$\left(\frac{gg'g'' \dots \times hh'h''}{b}\right) = \left(\frac{t}{b}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number.

Comparison of this result with the one obtained a moment ago shows that one always has:

$$(\gamma) \quad \left(\frac{t}{p}\right) = \left(\frac{t}{b}\right).$$

Let us now return to the equation $p = t^2 - bu^2$ and put it in congruence form:

$$t^2 \equiv bu^2 \pmod{p}.$$

From this, raising both sides to the power $\frac{p-1}{4}$, one obtains:

$$t^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p}.$$

The formula (α) , namely $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, where 2^ν denotes the highest power dividing u , can be presented in another manner. For this it is necessary to distinguish two cases, according as p is of the form $8n+1$ or of the form $8n+5$. If p is a prime number $8n+1$, one has, as is known, $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$; the formula in question therefore becomes in this case $\left(\frac{u}{p}\right) = 1$. If p is a prime number $8n+5$, the number ν is $= 1$; for if ν were greater than unity, u^2 would be divisible by 8, and $p = t^2 - bu^2$ would be of the form $8n+1$. Moreover, since in this case $\left(\frac{2}{p}\right) = -1$, formula (α) becomes $\left(\frac{u}{p}\right) = -1$. The two cases just examined are included in the formula $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$, which is equivalent to the congruence $u^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}} \pmod{p}$. Substituting the value which this gives

for $u^{\frac{p-1}{2}}$ in the congruence obtained above, one has:

$$t^{\frac{p-1}{2}} \equiv (-b)^{\frac{p-1}{4}} \pmod{p},$$

a result which shows that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to p . If one now compares this result with that contained in formula (γ) , one will arrive at the theorem which we shall state:

“Let b designate a prime number $4n + 3$, and p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$. Having written $p = t^2 - bu^2$ (where t is supposed odd), I say that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to b .”

4.

We shall now deduce from this theorem another by means of which one can decide still more quickly whether $-b$ is or is not a biquadratic residue with respect to p . Preserve the preceding notations and put $p = \varphi^2 + \psi^2$ (where ψ is supposed even). Equating this value of p to the one considered above, one has:

$$p = t^2 - bu^2 = \varphi^2 + \psi^2,$$

and, after transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + bu^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by b . We begin with the latter of these two cases. Let m be the greatest common divisor of φ and u , which will be odd (φ being odd) and not divisible by b , and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the last equation changes it into:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + bu'^2).$$

It is evident, from this equation, that $t + \psi$ is composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 + bu'^2$, and that the same is true of $t - \psi$. Denoting the factors of this last number by F and L , we shall have the equations:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + bu'^2 &= KL. \end{aligned}$$

It is very easy to make sure that the numbers E and F are relatively prime. Indeed, let δ be any simple factor of E (which will necessarily be odd, E itself being odd) and suppose that δ also divides F . Inspection of the preceding equations shows that, under this supposition, δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two. But on the other hand one sees that δ , as a prime divisor of E , divides m^2 , and consequently m and u (u being $= mu'$). The numbers t and u would therefore, under this supposition, have the common factor δ , which is absurd, since $p = t^2 - bu^2$ is a prime number. It is therefore proved that E and F cannot have a common divisor, and since the product of these numbers is a square, each of them is likewise a square. Thus one has:

$$\left(\frac{E}{b}\right) = 1, \quad \left(\frac{F}{b}\right) = 1.$$

As for the numbers K and L , they are likewise such that one has:

$$\left(\frac{K}{b}\right) = 1, \quad \left(\frac{L}{b}\right) = 1.$$

Indeed, one sees from the last equation (p. 74) that these numbers divide, the one and the other, the formula $\varphi'^2 + bu'^2$ (where φ' and bu' are relatively prime) and one knows by a known theorem that follows easily from the law of reciprocity (*Theory of Numbers*, no. 197) that every odd divisor R of such a formula is such that $\left(\frac{R}{b}\right) = 1$. Multiplying the first of the last relations by the third, and the second by the fourth, one obtains:

$$\left(\frac{EK}{b}\right) = 1, \quad \left(\frac{FL}{b}\right) = 1,$$

or, which is the same thing:

$$(\delta) \quad \left(\frac{t + \psi}{b}\right) = 1, \quad \left(\frac{t - \psi}{b}\right) = 1.$$

Examining in a similar way the case in which φ is divisible by b , one will find that, in this case, one of the numbers $t + \psi$ and $t - \psi$ is divisible by b , and that the other is, as in the case with which we have just been occupied, a quadratic residue with respect to b , so that, denoting by $t \pm \psi$ that one of these numbers which is divisible by b , one has:

$$(\delta') \quad t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

It follows from the theorem stated at the end of the preceding paragraph that, in order to decide whether $-b$ is or is not a biquadratic residue with respect to p , everything reduces to the question of knowing whether t is or is not a quadratic residue with respect to b . We can now show, by means of the results just obtained, that this latter question can be decided without knowing t , that is, without its being necessary to put p under the form $t^2 - bu^2$. Suppose first that φ is not divisible by b . The relations (δ) , which hold in this case, can be presented in this way, observing that b is a prime number $4n + 3$:

$$\left(\frac{\psi + t}{b}\right) = 1, \quad \left(\frac{\psi - t}{b}\right) = -1.$$

One immediately draws from the equation $p = t^2 - bu^2$: $t^2 \equiv p \pmod{b}$, a result which shows that, on solving the congruence $\chi^2 \equiv p \pmod{b}$, and denoting by χ one of its roots taken at random, one will have $t \equiv \chi$, or $t \equiv -\chi \pmod{b}$. In the first case, one has:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi}{b}\right), \quad \left(\frac{\chi + \psi}{b}\right) = 1,$$

for it is evident that the expressions $\left(\frac{t}{b}\right)$ and $\left(\frac{t+\psi}{b}\right)$ do not change when one puts, in place of t , another number χ which differs from t only by a multiple of b . Multiplying the last relations together, one obtains:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Consider now the other case, in which $t \equiv -\chi \pmod{b}$. One will then have:

$$\left(\frac{t}{b}\right) = -\left(\frac{\chi}{b}\right), \quad \left(\frac{\psi + \chi}{b}\right) = -1,$$

the last of these relations resulting from the formula $\left(\frac{\psi-t}{b}\right) = -1$, if one puts, in place of $-t$, the number χ , which differs from it only by a multiple of b . Multiplying the preceding relations together, one will find, as above:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

We have therefore arrived at this remarkable result: If φ is not divisible by b , one can decide very simply whether one has:

$$\left(\frac{t}{b}\right) = 1 \quad \text{or} \quad \left(\frac{t}{b}\right) = -1.$$

It will suffice to seek a number satisfying the congruence $\chi^2 \equiv p \pmod{b}$; having found such a number χ , one will always have:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Let us now come to the case in which φ is divisible by b . The relations (δ') which then hold are:

$$t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

In the last of these relations one may add to $t \mp \psi$ any multiple of b . If one adds to it the number $t \pm \psi$, which, by virtue of the first relation, is divisible by b , there will result $\left(\frac{2t}{b}\right) = 1$, a result which entails this other one: $\left(\frac{t}{b}\right) = \left(\frac{2}{b}\right)$.

Combining this result with a known theorem, one will see that, in the case of φ divisible by b , one has $\left(\frac{t}{b}\right) = 1$ or $\left(\frac{t}{b}\right) = -1$, according as t is of the form $8n + 7$ or of this one: $8n + 3$.

Comparing what has just been proved with the theorem established at the end of the preceding paragraph, one will see that it is possible to decide, independently of the knowledge of the number t , whether $-b$ is or is not a biquadratic residue with respect to p . Since the result thus reached no longer contains any trace of the number t , one is led to believe that it does not presuppose the possibility of the equation $t^2 - bu^2 = p$, from which the number t takes its origin, and that it is generally true for all values of b and p such that $\left(\frac{b}{p}\right) = 1$. This is indeed what takes place, as one can prove, by examining, instead of the equation $t^2 - bu^2 = p$, the more general equation $t^2 \pm Mu^2 = p$, where M denotes the product of any number of distinct prime numbers, and then combining the results of this examination with the following theorem, of whose truth one cannot doubt, but whose rigorous demonstration nevertheless presents difficulties:

“ c denoting any prime number whatever and p a prime number $4n + 1$, such that $\left(\frac{c}{p}\right) = 1$, one can always determine a number δ , composed of simple factors all smaller than c , and such that the equation $t^2 \pm c\delta u^2 = p$, is soluble.”

However that may be, one may remark that the theorem which we are about to state is rigorously proved, by what precedes, for all values

of b such that the formula $t^2 - bu^2$ has only the single quadratic divisor $\pm(t^2 - bu^2)$. Casting one's eyes over the first of the tables added to the *Theory of Numbers*, one sees that all the prime numbers $4n + 3$ smaller than 136 (the limit of the table) are in this case, excepting the single number 79. It would be easy, by following the path just indicated, to prove the truth of the theorem for this value or for any other particular value of b , whatever p may be; but in order to establish it in all its generality, it is necessary first to prove, as we have already said, that one can always satisfy the condition which we stated only a moment ago.

Theorem I.

" b denoting a prime number $4n + 3$, and p a prime number $4n + 1$ such that $\left(\frac{b}{p}\right) = 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one will have the following rule for deciding whether $-b$ is or is not a biquadratic residue with respect to p : If φ is divisible by b , $-b$ will or will not be a biquadratic residue, according as b is of the form $8n + 7$ or of this one: $8n + 3$. If φ is not divisible by b , one will seek a number χ such that $\chi^2 \equiv p \pmod{b}$. This being done, $-b$ will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{\chi(\chi+\psi)}{b}\right) = 1$ or $\left(\frac{\chi(\chi+\psi)}{b}\right) = -1$."

By successively taking $b = 3$, $b = 7$ etc., one will obtain the following particular theorems, which are analogous to the second of Mr. GAUSS's theorems and which must be regarded as rigorously proved by the considerations that we have set out in this paragraph and in the preceding one.

" p denoting a prime number $12n + 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one of the numbers φ and ψ will necessarily be divisible by 3. This being done, I say that -3 will or will not be a biquadratic residue with respect to p , according as it is ψ or φ which is divisible by 3."

" p denoting a prime number of one of these forms: $28n + 1$, $28n + 9$, $28n + 25$, if one puts $p = \varphi^2 + \psi^2$, I say that -7 will be a biquadratic residue with respect to p , if one of the numbers φ and ψ is divisible by 7, and that -7 will be a biquadratic non-residue with respect to p , if neither of these numbers is divisible by 7."

etc.

5.

Let us denote by a a prime number $4n + 1$, and by p a prime number likewise $4n + 1$, and moreover capable of being put under the form $t^2 - au^2$. We can therefore put $p = t^2 - au^2$ where the number t , as is easy to see, is necessarily odd. One will find, as in paragraph 3, that if one denotes by 2^ν the highest power of 2 that divides u , one has $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, a relation which one will then change, as in the cited place, into this one: $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$.

Let us now decompose the odd number t into its simple factors, putting $t = gg'g'' \dots \times hh'h'' \dots$, where the prime numbers $g, g', g'', \text{etc.}, h, h', h'', \text{etc.}$ are such that:

$$(\varepsilon) \quad \begin{cases} \left(\frac{-a}{g}\right) = 1, & \left(\frac{-a}{g'}\right) = 1, & \left(\frac{-a}{g''}\right) = 1, & \text{etc.} \\ \left(\frac{-a}{h}\right) = -1, & \left(\frac{-a}{h'}\right) = -1, & \left(\frac{-a}{h''}\right) = -1, & \text{etc.} \end{cases}$$

The equation $t^2 - au^2 = p$ gives immediately:

$$\begin{aligned} \left(\frac{-ap}{g}\right) &= 1, & \left(\frac{-ap}{g'}\right) &= 1, & \left(\frac{-ap}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{-ap}{h}\right) &= 1, & \left(\frac{-ap}{h'}\right) &= 1, & \left(\frac{-ap}{h''}\right) &= 1, & \text{etc.} \end{aligned}$$

Comparing these relations with the preceding ones, one obtains:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, & \left(\frac{p}{g'}\right) &= 1, & \left(\frac{p}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, & \left(\frac{p}{h'}\right) &= -1, & \left(\frac{p}{h''}\right) &= -1, & \text{etc.} \end{aligned}$$

The application of the law of reciprocity will then give, p being of the form $4n + 1$:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, & \left(\frac{g'}{p}\right) &= 1, & \left(\frac{g''}{p}\right) &= 1, & \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, & \left(\frac{h'}{p}\right) &= -1, & \left(\frac{h''}{p}\right) &= -1, & \text{etc.} \end{aligned}$$

whence one concludes, by multiplying:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign taking place according as the prime numbers h ,

$h', h'', \dots$ are even or odd in number. One can state this result in a slightly different way, by saying that $\left(\frac{t}{p}\right)$ is equal to the product of the second members of the relations (ε) .

In the relations (ε) , one may everywhere change $-a$ into a , provided that at the same time one changes the sign of the second members of those of these relations in which there occurs a prime number $g, g', g'', \dots, h, h', h'', \dots$ of the form $4n + 3$. One will thus have:

$$(\zeta) \quad \begin{cases} \left(\frac{a}{g}\right) = \pm 1, & \left(\frac{a}{g'}\right) = \pm 1, & \left(\frac{a}{g''}\right) = \pm 1, & \text{etc.} \\ \left(\frac{a}{h}\right) = \pm 1, & \left(\frac{a}{h'}\right) = \pm 1, & \left(\frac{a}{h''}\right) = \pm 1, & \text{etc.} \end{cases}$$

Comparing the second member of each of these relations with the second member of the corresponding relation of table (ε) , one will find as many changes of sign as there are prime numbers $4n + 3$ among the numbers $g, g', g'', \dots, h, h', h'', \dots$.

The number of changes will therefore be even, when among the numbers $g, g', g'', \dots, h, h', h'', \dots$ there is an even number of the form $4n + 3$. In this case, which will occur every time that $t = gg'g'' \dots \times hh'h'' \dots$ is of the form $4n + 1$, the product of the second members of the relations (ζ) will therefore be the same as the product of the second members of the relations (ε) .

On the other hand, one may, by virtue of the law of reciprocity, reverse the first members of the relations (ζ) without its being necessary to change the sign of any of the second members of these relations. The product of all the reversed expressions, that is, $\left(\frac{gg'g'' \dots \times hh'h'' \dots}{a}\right)$ or $\left(\frac{t}{a}\right)$, is therefore equal to the product of the second members of the relations (ζ) and consequently also, according to what has previously been seen, equal to the product of the second members of the relations (ε) . Now, having proved above that this last product is equal to $\left(\frac{t}{p}\right)$, one has:

$$\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right).$$

This result concerns the case in which t is of the form $4n + 1$; if one examines in a similar way the case of $t = 4n + 3$, one will find that one then has:

$$\left(\frac{t}{p}\right) = -\left(\frac{t}{a}\right).$$

These two cases are comprised in the following formula, to which we join a result obtained above:

$$(\eta) \quad \left(\frac{t}{p}\right) = (-1)^{\frac{t-1}{2}} \left(\frac{t}{a}\right), \quad \left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}.$$

The equation $t^2 - au^2 = p$ gives successively:

$$t^2 \equiv au^2, \quad t^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

whence one sees that a will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{t}{p}\right) = \left(\frac{u}{p}\right)$ or $\left(\frac{t}{p}\right) = -\left(\frac{u}{p}\right)$. Putting in place of $\left(\frac{t}{p}\right)$ and $\left(\frac{u}{p}\right)$ the values given by the expressions (η) , one can replace the preceding conditions by those that follow:

a will or will not be a biquadratic residue with respect to p , according as one has:

$$(\theta) \quad \left(\frac{t}{a}\right) = (-1)^{\frac{t-1}{2} + \frac{p-1}{4}} \quad \text{or} \quad \left(\frac{t}{a}\right) = -(-1)^{\frac{t-1}{2} + \frac{p-1}{4}},$$

a result that we shall not pause to demonstrate.

Let us now put $p = \varphi^2 + \psi^2$ (where ψ is supposed even) and equate this value of p to the one we considered previously. We shall thus have $p = t^2 - au^2 = \varphi^2 + \psi^2$, whence one concludes by transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + au^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by a . Let us begin with that in which φ is not divisible by a . Let m be the greatest common divisor of φ and of u , which will necessarily be odd and not divisible by a , and put $\varphi = m\varphi'$, $u = mu'$, values whose substitution in the equation obtained above changes it into this one:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + au'^2).$$

This equation shows that $t + \psi$ is composed of two factors, one of which divides m^2 , the other $\varphi'^2 + au'^2$, and that the same holds for $t - \psi$. If we denote the factors of $t + \psi$ by E and K and those of $t - \psi$ by F and L , we shall have:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + au'^2 &= KL. \end{aligned}$$

One will ascertain, as in paragraph 4, that the numbers E and F are relatively prime. It follows from this and from the equation $m^2 = EF$ that each of these numbers is a square, so that:

$$\left(\frac{E}{a}\right) = 1, \quad \left(\frac{F}{a}\right) = 1.$$

The difference of the odd numbers $t + \psi$ and $t - \psi$ being 2ψ and consequently divisible by 4, one sees that these two numbers are either both of the form $4n + 1$, or both of the form $4n + 3$. Since, on the other hand, the numbers E and F are both of the form $4n + 1$ (these numbers being odd squares), it follows from the equations $t + \psi = EK$, $t - \psi = FL$, that the numbers K and L are either both of the form $4n + 1$ or both of the form $4n + 3$, and that the first or the second of these cases holds according as the numbers $t + \psi$ and $t - \psi$ are both of the form $4n + 1$, or both of the form $4n + 3$. Let us now distinguish two cases, according as p is of the form $8n + 1$ or of this one: $8n + 5$. One sees, from the equation $p^2 = \varphi^2 + \psi^2$, that, in the first case, ψ is divisible by 4, whence it follows that $t + \psi$ and $t - \psi$ are then both of the form $4n + 1$ or both of the form $4n + 3$, according as t is of the form $4n + 1$, or of this one: $4n + 3$. The contrary holds in the case of $p = 8n + 5$, and the results relating to these two cases can be comprised in this statement:

The numbers $t + \psi$ and $t - \psi$, and consequently also the numbers K and L , are both of the form $4n + 1$ or of this one: $4n + 3$, according as $(-1)^{\frac{p-1}{4} + \frac{t-1}{2}}$ is equal to 1 or to -1 .

It follows from the equation $\varphi'^2 + au'^2 = KL$ that the numbers K and L are divisors of the formula $\varphi'^2 + au'^2$ (where φ' and au' are relatively prime and where a denotes a prime number $4n + 1$). Now, one knows that, if R denotes an odd divisor of such a formula, one has:

$$\left(\frac{R}{a}\right) = 1, \quad \text{or} \quad \left(\frac{R}{a}\right) = -1,$$

according as R is of the form $4n + 1$ or of the form $4n + 3$. (*Theory of Numbers*, no. 196.)

Applying this theorem to the numbers K and L , one obtains:

$$\left(\frac{K}{a}\right) = \left(\frac{L}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

whence one will then conclude, by multiplying by the expressions $\left(\frac{E}{a}\right) = 1$, $\left(\frac{F}{a}\right) = 1$, found above:

$$\left(\frac{t + \psi}{a}\right) = \left(\frac{t - \psi}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

or, which comes to the same thing, a being a prime number $4n + 1$:

$$\left(\frac{\psi + t}{a}\right) = \left(\frac{\psi - t}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}}.$$

If one compares what has just been proved with the result (θ) , obtained above, one will see that a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right) = \left(\frac{t}{a}\right),$$

or:

$$\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right) = -\left(\frac{t}{a}\right).$$

The equation $p = t^2 - au^2$ shows that, if one denotes by χ either of the two roots of the congruence $\chi^2 \equiv p \pmod{a}$, one has $t \equiv \chi$ or $t \equiv -\chi \pmod{a}$. It follows from this, a being of the form $4n+1$, that $\left(\frac{t}{a}\right) = \left(\frac{\chi}{a}\right)$. On the other hand, since by virtue of what precedes $\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right)$, one sees that one always has:

$$\left(\frac{\psi+\chi}{a}\right) = \left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right).$$

One concludes from this, having regard to what was said only a moment ago, that the number a will be a biquadratic residue with respect to p when one has $\left(\frac{\chi+\psi}{a}\right) = \left(\frac{\chi}{a}\right)$ or, which is the same thing, when $\left(\frac{\chi(\chi+\psi)}{a}\right) = 1$, and that a will be a biquadratic non-residue with respect to p when one has $\left(\frac{\chi+\psi}{a}\right) = -\left(\frac{\chi}{a}\right)$, or, which comes to the same thing, when $\left(\frac{\chi(\chi+\psi)}{a}\right) = -1$.

This result concerns the case in which φ is not divisible by a . It remains to examine the case of φ divisible by a . Treating this case in a similar way, one finds that, when it holds, one of the numbers $t+\psi$, $t-\psi$ (we shall denote it by $t \pm \psi$) is divisible by a and that, as in the first case, the other, $t \mp \psi$, is such that:

$$\left(\frac{t \mp \psi}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

or, which comes to the same thing, on adding to $t \mp \psi$ the number $t \pm \psi$, a multiple of a :

$$\left(\frac{2t}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}}.$$

Comparing this with what was proved above (θ) , one will find that, in the case of φ divisible by a , a is or is not a biquadratic residue with respect to p , according as one has $\left(\frac{2}{a}\right) = 1$, or $\left(\frac{2}{a}\right) = -1$, or, which is the

same thing, by a known theorem, according as a is of the form $8n + 1$ or of the form $8n + 5$.

The results we have just obtained stand as do those that we established in paragraph 4; they still hold even when p cannot be put under the form $t^2 - au^2$, and presuppose only that the prime numbers $a = 4n + 1$, and $p = 4n + 1$, are such that $\left(\frac{a}{p}\right) = 1$. To prove them in this extension, it will suffice to follow the procedure indicated towards the end of paragraph 4, and to rely on the auxiliary proposition discussed there, whose statement was given the generality necessary for it to serve as a basis, at the same time, for the demonstration of the general theorem at the end of paragraph 4 and for that of the theorem we are about to state. These two theorems properly form only one, and by proceeding as was said in the cited place, that is, by applying to the equation $t^2 \pm Mu^2 = p$, where M denotes the product of any number of distinct prime numbers, considerations analogous to those employed in this paragraph and in the preceding ones, one will demonstrate these two theorems simultaneously, or rather one will arrive at a theorem in whose statement they are united.

Theorem II.

“ a denoting a prime number $4n + 1$, and p another prime number $4n + 1$, such that $\left(\frac{a}{p}\right) = 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even), one will be able to decide in the following way whether a is or is not a biquadratic residue with respect to p . If φ is divisible by a , a will or will not be a biquadratic residue with respect to p , according as a is of the form $8n + 1$ or of this one: $8n + 5$. If φ is not divisible by a , one will seek a number χ such that $\chi^2 \equiv p \pmod{a}$. This being done, a will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{\chi(\chi+\psi)}{a}\right) = 1$ or $\left(\frac{\chi(\chi+\psi)}{a}\right) = -1$.”

By successively taking $a = 5$, $a = 13$, one will have the following particular theorems, which must be regarded as rigorously proved by what precedes, the formula $t^2 - au^2$ having, for these values of a , no quadratic divisor other than this one: $t^2 - au^2$, or in other words, every divisor of $t^2 - au^2$ being itself of the form $t^2 - au^2$.

“ p denoting a prime number of one of the forms $20n + 1$, $20n + 9$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one readily ascertains that one of the numbers φ , ψ is a multiple of 5. This being done, I say that 5 will or will not be a biquadratic residue with respect to p , according as it is ψ or φ that is divisible by 5.”

etc.

In indicating above a method that could serve to establish theorems I and II in all their generality, we said that, in following this method, one found oneself obliged to rely on a subsidiary proposition that appears rather difficult to demonstrate with complete rigour. On reflecting on this difficulty, I recognized that it was very easy to remedy it by modifying slightly the procedure outlined above. This modification is extremely slight and consists in this: instead of considering the equations $t^2 - bu^2 = p$, $t^2 - au^2 = p$, which may fail to be possible although one has $\left(\frac{b}{p}\right) = 1$, $\left(\frac{a}{p}\right) = 1$, one must apply considerations of the preceding kind to the more general equations $t^2 - bu^2 = ps^2$, $t^2 - au^2 = ps^2$ (one advantageously replaces the latter by this one: $t^2 + au^2 = ps$, which can be treated more simply), which can always be satisfied when the conditions $\left(\frac{b}{p}\right) = 1$, $\left(\frac{a}{p}\right) = 1$ hold, as follows from the beautiful theorem that Mr. LEGENDRE gave for deciding the possibility or impossibility of the equation $\alpha x^2 + \beta y^2 = \gamma z^2$ (*Theory of Numbers*, no. 27).

The demonstration thus modified, besides being entirely rigorous, has the further advantage of greater simplicity, as any reader who has fully grasped the spirit of the preceding considerations will be able to judge by developing this demonstration according to the indication just given.

Addition to the preceding memoir.

Although the indication that we gave at the end of the preceding memoir concerning the procedure to follow in order to establish in an entirely rigorous way theorems I and II of our memoir may suffice for any reader familiar with indeterminate analysis, we believe we are not doing something useless by developing in detail, in this addition, the demonstrations of the theorems in question, these demonstrations being susceptible of being greatly

simplified by means of a few considerations that may not immediately present themselves to the mind. We begin with the demonstration of the first of the theorems cited.

Let us denote by p a prime number $4n + 1$, by b a prime number $4n + 3$, such that $\left(\frac{b}{p}\right) = 1$, and consider the equation:

$$(\alpha') \quad t^2 - bu^2 = ps^2.$$

By the theorem already cited (*Theory of Numbers*, no. 27), the necessary conditions for this equation to be possible are these: $\left(\frac{b}{p}\right) = 1$ and $\left(\frac{p}{b}\right) = 1$. Now, of these conditions the first holds by hypothesis, and as for the second, it follows from the theorem known as the law of reciprocity that it always holds when the first is satisfied. The equation is therefore soluble. It is evident that one may suppose the numbers t, u, s that satisfy it to be relatively prime when compared two at a time. The equation being in this state, one of these three numbers will be even, the other two odd, and one readily ascertains that s cannot be even. The number s being odd, of the numbers t, u , one will be even, the other odd, and there would be two cases to examine according as t is even or odd. But although these two cases can be treated by the same analysis, it is simpler to consider only one of them, that of t odd for example, and to show that the other can easily be reduced to it. Let us therefore show that it is always permissible to suppose t odd in the equation $t^2 - bu^2 = ps^2$. If t were even in the equation (α') , one would deduce from the numbers t, u , by the means indicated in the note relating to the beginning of paragraph 3 of the preceding memoir, other numbers t' (odd), u' (even) such that $t'^2 - bu'^2 = ps^2$. We may therefore regard t as odd in all that follows.

Let us decompose the numbers t and u into their simple factors, putting:

$$t = ll'l'' \dots, \quad u = 2^\nu kk'k'' \dots,$$

$l, l', l'', \dots$ and $k, k', k'', \dots$ denoting odd prime numbers. Mere inspection of the equation (α') gives:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \dots$$

One will then have, on applying the law of reciprocity, p being a prime number $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \dots$$

Taking the product of all these expressions and of the identical formula $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

The number p is of one of the two forms $8n + 1$, $8n + 5$. If p is of the first form, one has, by a known theorem, $\left(\frac{2}{p}\right) = 1$, and consequently also $\left(\frac{2^\nu}{p}\right) = 1$. Thus $\left(\frac{u}{p}\right) = 1$. If p is of the form $8n + 5$, the number u will be oddly even, so that $\nu = 1$, and since one has, on the other hand, for prime numbers $p = 8n + 5$, $\left(\frac{2}{p}\right) = -1$, one obtains $\left(\frac{u}{p}\right) = -1$. The two cases just considered are comprised in the formula:

$$(\beta') \quad \left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}.$$

Let us return to the equation (α') from which we immediately obtain:

$$\left(\frac{-b}{l}\right) = \left(\frac{p}{l}\right), \quad \left(\frac{-b}{l'}\right) = \left(\frac{p}{l'}\right), \quad \left(\frac{-b}{l''}\right) = \left(\frac{p}{l''}\right), \quad \dots$$

Since p is a prime number $4n + 1$, and b a prime number $4n + 3$, one will find, on applying the law of reciprocity:

$$\left(\frac{l}{b}\right) = \left(\frac{l}{p}\right), \quad \left(\frac{l'}{b}\right) = \left(\frac{l'}{p}\right), \quad \left(\frac{l''}{b}\right) = \left(\frac{l''}{p}\right), \quad \dots$$

Multiplication will then give:

$$\left(\frac{ll'l'' \dots}{b}\right) = \left(\frac{ll'l'' \dots}{p}\right),$$

or, which comes to the same thing:

$$(\gamma') \quad \left(\frac{t}{b}\right) = \left(\frac{t}{p}\right).$$

Putting the equation (α') into the form of a congruence, one has:

$$t^2 \equiv bu^2 \pmod{p}.$$

One concludes from this, by raising both members to the power $\frac{p-1}{4}$:

$$t^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

or, which comes to the same thing, by replacing $u^{\frac{p-1}{2}}$ by its value $(-1)^{\frac{p-1}{4}}$ given by the formula (β') :

$$t^{\frac{p-1}{2}} \equiv (-b)^{\frac{p-1}{4}} \pmod{p}.$$

This last congruence shows that $-b$ is or is not a biquadratic residue with respect to p , according as one has $\left(\frac{t}{p}\right) = 1$, or $\left(\frac{t}{p}\right) = -1$. If one compares this result with the formula (γ') , one will obtain this statement:

$$(\delta') \quad \begin{array}{l} -b \text{ is or is not a biquadratic residue with respect to } p, \text{ according as one has } \left(\frac{t}{b}\right) = 1 \\ \text{or } \left(\frac{t}{b}\right) = -1. \end{array}$$

Let us now decompose the prime number p into two squares, putting $p = \varphi^2 + \psi^2$ (where ψ is supposed even) and substitute this value of p in the equation (α') . One thus obtains:

$$t^2 - bu^2 = s^2\varphi^2 + s^2\psi^2,$$

and one will then have, on transposing:

$$t^2 - \psi^2 s^2 = (t + \psi s)(t - \psi s) = s^2\varphi^2 + bu^2.$$

The numbers s, t, u being relatively prime two at a time, one sees from the equation (α') that s cannot be divisible by b . We shall now distinguish two cases according as φ is or is not divisible by b , and we shall first examine the latter of these two cases. If we denote by m the greatest common divisor of φ and u , m will be odd and not divisible by b , like φ . Let us put $\varphi = m\varphi', u = mu'$ and substitute these values in the last equation. One thus obtains:

$$(t + \psi s)(t - \psi s) = m^2(s^2\varphi'^2 + bu'^2).$$

Since $s\varphi'$ and bu' are relatively prime, by what precedes, it follows from a known theorem, a very simple consequence of the law of reciprocity (*Theory of Numbers* no. 197), that every divisor R of the binomial factor of the second member, which is evidently odd, is such that $\left(\frac{R}{b}\right) = 1$.

The last equation shows that each of the numbers $t + \psi s, t - \psi s$ is composed of two factors, one a divisor of m^2 , the other a divisor of $s^2\varphi'^2 + bu'^2$. If we denote these 4 factors by E, F, K and L , we shall have the equations:

$$\begin{array}{l} t + \psi s = EK, \quad m^2 = EF, \\ t - \psi s = FL, \quad \varphi'^2 s^2 + bu'^2 = KL. \end{array}$$

It is easy to ascertain that the numbers E and F , which are both odd, as divisors of m^2 , are relatively prime. Indeed, if these numbers were both divisible by the same prime number δ , δ would divide each of the numbers $t + \psi s, t - \psi s$ and consequently also their half-sum and their half-difference, that is to say, the numbers t and ψs . On the other

hand, since δ divides m^2 , it will also divide the number φ , φ being equal to $m\varphi'$; it follows from this and from $\varphi^2 + \psi^2$ being equal to the prime number p , that ψ is not divisible by b . It would therefore follow, by what precedes, that δ was a common divisor of t and s , which is impossible, these numbers being relatively prime. Thus the numbers E and F are relatively prime, and since their product is a square, each of them will likewise be a square. One therefore has:

$$\left(\frac{E}{b}\right) = 1, \quad \left(\frac{F}{b}\right) = 1.$$

One also has:

$$\left(\frac{K}{b}\right) = 1, \quad \left(\frac{L}{b}\right) = 1,$$

K and L being divisors of $s^2\varphi'^2 + bu'^2$. One concludes from this, by multiplying:

$$\left(\frac{EK}{b}\right) = 1, \quad \left(\frac{FL}{b}\right) = 1,$$

or, which is the same thing:

$$\left(\frac{t + \psi s}{b}\right) = 1, \quad \left(\frac{t - \psi s}{b}\right) = 1.$$

If one observes that b is a prime number $4n + 3$, one will be able to present the last of these relations in this other manner:

$$\left(\frac{\psi s - t}{b}\right) = -1.$$

This being so, one sees that both can be comprised in this formula:

$$(\varepsilon') \quad \left(\frac{\psi s \pm t}{b}\right) = \pm 1,$$

where the signs are at will, but the same in both members.

Since one has $\left(\frac{p}{b}\right) = 1$, the congruence $\chi^2 \equiv p \pmod{b}$ is possible and has, as one knows, two roots between the limits $-\frac{1}{2}b$ and $\frac{1}{2}b$. Let us denote by χ one of these roots taken at random. The equation (α') gives this congruence: $t^2 \equiv ps^2 \pmod{b}$. If one compares this with the preceding one, one will have $t^2 \equiv \chi^2 s^2 \pmod{b}$. One concludes from this that $\pm t \equiv \chi s \pmod{b}$, the sign being sometimes positive, sometimes negative and depending on the choice made between the two roots of the congruence $\chi^2 \equiv p \pmod{b}$. The signs being at will in the formula (ε') , one may suppose that they are the same there as in this one: $\pm t \equiv \chi s \pmod{b}$. Replacing $\pm t$ by the number χs , which differs from $\pm t$ only by a multiple of b , one obtains:

$$\left(\frac{\psi s + \chi s}{b}\right) = \pm 1.$$

On the other hand, if one recalls that b is of the form $4n + 3$, one will obtain from the formula $\pm t \equiv \chi s \pmod{b}$: $\pm \left(\frac{t}{b}\right) = \left(\frac{\chi s}{b}\right)$. If one multiplies this equality by the preceding one, one will have, the signs being the same in these two formulae:

$$\left(\frac{\chi(\chi + \psi)}{b}\right) \left(\frac{s^2}{b}\right) = \left(\frac{t}{b}\right)$$

or, which comes to the same thing:

$$\left(\frac{\chi(\chi + \psi)}{b}\right) = \left(\frac{t}{b}\right).$$

If one compares this formula with what was proved above (δ'), one will obtain this result:

$$(\zeta') \quad \begin{array}{l} -b \text{ is or is not a biquadratic residue with respect to } p, \\ \text{according as one has } \left(\frac{\chi(\chi + \psi)}{b}\right) = 1 \text{ or } \left(\frac{\chi(\chi + \psi)}{b}\right) = -1. \end{array}$$

This proposition concerns the case where φ is not divisible by b . Let us now examine the case of φ divisible by b . Let b^{1+k} and b^h be respectively the highest powers of b that divide φ and u (k and h may be zero), and let m moreover be the greatest common divisor of $\frac{\varphi}{b^{1+k}}$ and $\frac{u}{b^h}$. If we put $\varphi = b^{1+k}m\varphi'$, $u = b^hmu'$, the numbers φ' and u' , of which the first is odd, the second even, will be relatively prime, and not divisible by b , and m will be odd and not divisible by b . Substitution of the preceding values gives this equation:

$$(t + \psi s)(t - \psi s) = m^2[b^{2k+2}(s\varphi')^2 + b^{2h+1}u'^2].$$

We must now distinguish two cases according as one has $k \geq h$ or $k < h$. In the first, the binomial factor of the second member may be put in this form:

$$b^{2h+1}[u'^2 + b(b^{k-h}s\varphi')^2],$$

where it must be observed that the numbers u' and $b^{k-h}s\varphi'$ are relatively prime. In the case of $k < h$, one puts the binomial factor in the following form:

$$b^{2k+2}[(s\varphi')^2 + b(b^{h-k-1}u')^2],$$

where the numbers $s\varphi'$ and $b^{h-k-1}u'$ are relatively prime. One therefore sees that the binomial factor is always the product of a power of b and an odd expression such as $g^2 + bh^2$, in which g and bh are relatively prime. Denoting by b^r the power of b in question, one may therefore write the last

equation in this manner:

$$(t + \psi s)(t - \psi s) = m^2 b^r (g^2 + bh^2).$$

The numbers $t + \psi s$, $t - \psi s$ cannot both be divisible by b , for then their half-sum, which is t , would also be so, which was seen above to be impossible. If one denotes by $t \pm \psi s$ that one of these two numbers which is divisible by b , one has $t \pm \psi s \equiv 0 \pmod{b}$ and I say that the other $t \mp \psi s$ is such that $\left(\frac{t \mp \psi s}{b}\right) = 1$. Indeed $t \mp \psi s$ is composed of two factors E and K of which the first E divides m^2 , the second $g^2 + bh^2$.

Now, one ascertains, as in the case of φ not divisible by b , that E is a square, so that one has $\left(\frac{E}{b}\right) = 1$. One also has $\left(\frac{K}{b}\right) = 1$, K being a divisor of $g^2 + bh^2$. One concludes from this by multiplying:

$$\left(\frac{EK}{b}\right) = \left(\frac{t \mp \psi s}{b}\right) = 1.$$

It is permissible to increase $t \mp \psi s$ in this formula by any multiple of b . Adding the number $t \pm \psi s$, which was seen to be such a multiple, one obtains $\left(\frac{2t}{b}\right) = 1$ or, which comes to the same thing: $\left(\frac{t}{b}\right) = \left(\frac{2}{b}\right)$, a result whose comparison with a known theorem shows that in the case of φ divisible by b , one has $\left(\frac{t}{b}\right) = 1$ or $\left(\frac{t}{b}\right) = -1$, according as b is of the form $8n + 7$ or of this one: $8n + 3$. If one combines this with the result obtained above (δ'), one finds that, when φ is a multiple of b :

$-b$ is or is not a biquadratic residue with respect to p , according as b is of the form $8n + 7$ or of this one: $8n + 3$.

Combining this last result with that reached above (ζ'), one obtains Theorem I stated at the end of paragraph 4 of the preceding memoir, a theorem thus established in a very simple and entirely rigorous manner.

Let us now turn to the demonstration of Theorem II. Let us denote, as in all that precedes, by a a prime number $4n + 1$ and by p another prime number likewise of the form $4n + 1$ and such that $\left(\frac{a}{p}\right) = 1$. One concludes from this, on applying the law of reciprocity, that one also has $\left(\frac{p}{a}\right) = 1$. These conditions holding, it follows from the theorem already cited several

times, that the equation:

$$(\eta') \quad t^2 + au^2 = ps^2,$$

can be solved. If one supposes that the numbers t , u , s , compared two at a time, have no common divisor, s will be odd and of the other two one will be even, the other odd, as is very easy to see. It is moreover evident that, if the equation is in this state, u is not divisible by p , s is not divisible by a , and t is divisible neither by a nor by p .

Let us decompose the numbers t and u into their simple factors, putting $u = 2^\nu k k' k'' \dots$, $t = 2^\mu l l' l'' \dots$, k , k' , $k'' \dots$ and l , l' , $l'' \dots$ denoting odd prime numbers, and one of the numbers μ , ν being equal to zero.

The equation (η') first gives:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \dots$$

whence one then concludes, on applying the law of reciprocity:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \dots$$

Taking the product of the preceding equations and of the identical equation $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

It also follows from the equation (η') , that one has:

$$\left(\frac{a}{l}\right) = \left(\frac{p}{l}\right), \quad \left(\frac{a}{l'}\right) = \left(\frac{p}{l'}\right), \quad \left(\frac{a}{l''}\right) = \left(\frac{p}{l''}\right), \quad \dots$$

Since the prime numbers a and p are both of the form $4n + 1$, one concludes from this, by the law of reciprocity:

$$\left(\frac{l}{a}\right) = \left(\frac{l}{p}\right), \quad \left(\frac{l'}{a}\right) = \left(\frac{l'}{p}\right), \quad \left(\frac{l''}{a}\right) = \left(\frac{l''}{p}\right), \quad \dots$$

Multiplying all these relations together, one obtains the equation:

$$\left(\frac{l l' l'' \dots}{a}\right) = \left(\frac{l l' l'' \dots}{p}\right),$$

which, if one multiplies both its members by $\left(\frac{2^\mu}{a}\right) \left(\frac{2^\mu}{p}\right)$, changes into this one:

$$\left(\frac{t}{a}\right) \left(\frac{2^\mu}{p}\right) = \left(\frac{t}{p}\right) \left(\frac{2^\mu}{a}\right).$$

Let us now distinguish two cases, according as u is even or odd. If

u is even, t is odd; one then has $\mu = 0$, and the last equality reduces to this one:

$$\left(\frac{t}{a}\right) = \left(\frac{t}{p}\right).$$

As for the relation $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, it is easy to see that it is equivalent, in this case, to this one:

$$\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}.$$

To be convinced of this, it suffices to observe that, if p is of the form $8n+1$, one has $\left(\frac{2}{p}\right) = 1$, and consequently also $\left(\frac{2^\nu}{p}\right) = 1$, and that, if p has the form $8n+5$, ν is equal to unity and one has for every prime number $p = 8n+5$, $\left(\frac{2}{p}\right) = -1$.

Let us pass to the other case, that of t even and u odd. One then has $\nu = 0$, which reduces the equation $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$ to $\left(\frac{u}{p}\right) = 1$.

If the numbers p and a are both of the form $8n+1$, or both of the form $8n+5$, one has $\left(\frac{2}{a}\right) = \left(\frac{2}{p}\right)$ and consequently also $\left(\frac{2^\mu}{a}\right) = \left(\frac{2^\mu}{p}\right)$, and the equality $\left(\frac{t}{p}\right)\left(\frac{2^\mu}{a}\right) = \left(\frac{t}{a}\right)\left(\frac{2^\mu}{p}\right)$ changes into this one: $\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right)$. If, on the contrary, of the numbers a, p , one is of the form $8n+1$, the other of the form $8n+5$, one has $\left(\frac{2}{a}\right) = -\left(\frac{2}{p}\right)$. On the other hand, the equation (η') shows that, in this case, the number t is oddly even, so that $\mu = 1$. One therefore then has $\left(\frac{t}{p}\right) = -\left(\frac{t}{a}\right)$. These two results are comprised in the formula:

$$\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right) (-1)^{\frac{a-1}{4} + \frac{p-1}{4}}.$$

If we combine all that we have just proved, we shall obtain this statement:

If u is even, one has $\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right)$, $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$; if u is odd, one has $\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right) (-1)^{\frac{p-1}{4} + \frac{a-1}{4}}$, $\left(\frac{u}{p}\right) = 1$.

Comparing this result with the congruence $t^{\frac{p-1}{2}} \equiv (-a)^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p}$, which follows immediately from the equation (η') , one obtains this theorem:

If u is even, a is or is not a biquadratic residue with respect to p , according as one has:

$$(\vartheta') \quad \left(\frac{t}{a}\right) = 1 \quad \text{or} \quad \left(\frac{t}{a}\right) = -1;$$

if u is odd, a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{t}{a}\right) = (-1)^{\frac{a-1}{4}} \quad \text{or} \quad \left(\frac{t}{a}\right) = -(-1)^{\frac{a-1}{4}}.$$

Let us decompose p into two squares, putting $p = \varphi^2 + \psi^2$ (ψ being supposed even). Substitution of this value in the equation (η') gives:

$$(t + \psi s)(t - \psi s) = \varphi^2 s^2 - au^2.$$

We shall now distinguish two cases according as φ is or is not divisible by a .

First case: φ is not divisible by a .

Since φ is odd and not divisible by a , the greatest common divisor of φ and of u , which we shall denote by m , will also be odd and not divisible by a . If we put $\varphi = m\varphi'$, $u = mu'$, the preceding equation changes into this one:

$$(t + \psi s)(t - \psi s) = m^2(\varphi'^2 s^2 - au'^2);$$

s is prime to au and consequently also to au' , φ' is likewise prime to au' ; thus $\varphi's$ and au' have no common divisor. This being so, it follows from a known theorem (*Theory of Numbers* no. 197) that every odd divisor R of $s^2\varphi'^2 - au'^2$ is such that one has $\left(\frac{R}{a}\right) = 1$.

Denoting four indeterminates by E , K , F and L , we may replace the last equation by those which I shall write:

$$\begin{aligned} t + \psi s &= EK, & m^2 &= EF, \\ t - \psi s &= FL, & (s\varphi')^2 - au'^2 &= KL. \end{aligned}$$

One readily ascertains that the odd numbers E and F are relatively prime. It follows from this that each of them is a square, so that one has:

$$\left(\frac{E}{a}\right) = 1, \quad \left(\frac{F}{a}\right) = 1.$$

We must now consider successively the case of u even and that of u odd. If u is even, $u' = \frac{u}{m}$ will likewise be so, and since $s\varphi'$ is always odd, $(s\varphi')^2 - au'^2$ will also be odd. The numbers K and L will therefore

both be odd in this case, and since they divide the preceding binomial, one has:

$$\left(\frac{K}{a}\right) = 1, \quad \left(\frac{L}{a}\right) = 1.$$

Multiplying these relations by those which we have just written, one obtains:

$$\left(\frac{EK}{a}\right) = \left(\frac{t + \psi s}{a}\right) = 1, \quad \left(\frac{FL}{a}\right) = \left(\frac{t - \psi s}{a}\right) = 1.$$

Since a is a prime number $4n + 1$, it is permissible to write the formula $\left(\frac{t - \psi s}{a}\right) = 1$ in this other manner: $\left(\frac{\psi s - t}{a}\right) = 1$. One therefore has, in the case of u even, $\left(\frac{\psi s \pm t}{a}\right) = 1$, the double sign being at will.

Let us pass to the case where u is odd. The number $u' = \frac{u}{m}$ will likewise be odd in this case, and since the odd squares $(s\varphi')^2$, u'^2 are of the form $8n + 1$, one sees that the binomial $(s\varphi')^2 - au'^2$ will be of the form $8n$ or of the form $8n + 4$, according as a is of the form $8n + 1$ or of this one: $8n + 5$. It is easy to ascertain that, if a is of the form $8n + 1$, one has $\left(\frac{K}{a}\right) = 1$, $\left(\frac{L}{a}\right) = 1$. Indeed, if one puts $K = 2^\beta K'$, 2^β being the highest power of 2 that divides K , K' will be an odd divisor of $(s\varphi')^2 - au'^2$ and one has $\left(\frac{K'}{a}\right) = 1$. On the other hand, one knows, by a known theorem, that every prime number $a = 8n + 1$, is such that $\left(\frac{2}{a}\right) = 1$. One concludes from this that $\left(\frac{2^\beta}{a}\right) = 1$, and then multiplying by $\left(\frac{K'}{a}\right) = 1$, one obtains $\left(\frac{K}{a}\right) = 1$. In an entirely similar manner one proves that one also has $\left(\frac{L}{a}\right) = 1$. Multiplying these relations by the following: $\left(\frac{E}{a}\right) = 1$, $\left(\frac{F}{a}\right) = 1$, obtained above, one obtains:

$$\left(\frac{EK}{a}\right) = \left(\frac{t + \psi s}{a}\right) = 1, \quad \left(\frac{FL}{a}\right) = \left(\frac{t - \psi s}{a}\right) = 1.$$

It is easy to see that these results, which hold when u is odd and at the same time a is of the form $8n + 1$, can be combined in the formula $\left(\frac{\psi s \pm t}{a}\right) = 1$, where the double sign is at will.

The number u still being odd, let us suppose a of the form $8n + 5$. The product KL , which is equivalent to the binomial $(s\varphi')^2 - au'^2$, then being of the form $8n + 4$, and the numbers K and L being evidently both even, each

of them will be of the form $4n + 2$, so that, putting $K = 2K'$, $L = 2L'$, K' and L' will be odd; and since these latter numbers are moreover divisors of $(s\varphi')^2 - au'^2$, it follows from what was said above, that one has $\left(\frac{K'}{a}\right) = 1$, $\left(\frac{L'}{a}\right) = 1$. But one has, on the other hand, a being of the form $8n + 5$, $\left(\frac{2}{a}\right) = -1$. Multiplying by the preceding equalities, one first obtains $\left(\frac{K}{a}\right) = -1$, $\left(\frac{L}{a}\right) = -1$, and if one then multiplies by the equalities $\left(\frac{E}{a}\right) = 1$, $\left(\frac{F}{a}\right) = 1$, found above, one has:

$$\left(\frac{EK}{a}\right) = \left(\frac{t + \psi s}{a}\right) = -1, \quad \left(\frac{FL}{a}\right) = \left(\frac{t - \psi s}{a}\right) = -1.$$

These results, which concern the case where u is odd and the number a is of the form $8n + 5$, are comprised in the double formula:

$$\left(\frac{\psi s \pm t}{a}\right) = -1.$$

Summing up all that precedes, one obtains this statement:

“If u is even, one has $\left(\frac{\psi s \pm t}{a}\right) = 1$; if, on the contrary, u is odd, one has $\left(\frac{\psi s \pm t}{a}\right) = (-1)^{\frac{a-1}{4}}$, the double sign being at will in both cases.”

It was seen above (ϑ') that, when u is even, a is or is not a biquadratic residue with respect to p , according as one has $\left(\frac{t}{a}\right) = 1$, or $\left(\frac{t}{a}\right) = -1$. On the other hand, it follows from what precedes, that one has, in this same case, $\left(\frac{\psi s \pm t}{a}\right) = 1$. One concludes from this that, if u is even, a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{t}{a}\right) = \left(\frac{\psi s \pm t}{a}\right) \quad \text{or} \quad \left(\frac{t}{a}\right) = -\left(\frac{\psi s \pm t}{a}\right).$$

It was also seen above (ϑ') that, when u is odd, a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{t}{a}\right) = (-1)^{\frac{a-1}{4}} \quad \text{or} \quad \left(\frac{t}{a}\right) = -(-1)^{\frac{a-1}{4}}.$$

On the other hand, the preceding statement shows, that one has, in the case of u odd, $\left(\frac{\psi s \pm t}{a}\right) = (-1)^{\frac{a-1}{4}}$. Comparing these two results, one concludes that, when

u is odd, a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{t}{a}\right) = \left(\frac{\psi s \pm t}{a}\right) \quad \text{or} \quad \left(\frac{t}{a}\right) = -\left(\frac{\psi s \pm t}{a}\right).$$

The conclusion being the same in the case of u even and in that of u odd, one may state this theorem:

“If a is a biquadratic residue with respect to p , one has $\left(\frac{t}{a}\right) = \left(\frac{\psi s \pm t}{a}\right)$; if, on the contrary, a is not a biquadratic residue with respect to p , one has $\left(\frac{t}{a}\right) = -\left(\frac{\psi s \pm t}{a}\right)$, the double sign being at will in both cases.”

Since one has $\left(\frac{p}{a}\right) = 1$, the congruence $\chi^2 \equiv p \pmod{a}$ will be possible and will have, as one knows, two roots between the limits $-\frac{1}{2}a$ and $\frac{1}{2}a$. Let us suppose that χ denotes either of these roots indifferently. One immediately obtains from the equation (η') : $t^2 \equiv ps^2 \pmod{a}$, a congruence whose comparison with the preceding one gives $t^2 \equiv \chi^2 s^2$, $\pm t \equiv \chi s \pmod{a}$, the upper sign holding for one and the lower sign for the other of the two roots of the congruence $\chi^2 \equiv p \pmod{a}$. If a is a biquadratic residue with respect to p , one has, as was seen just a moment ago, $\left(\frac{t}{a}\right) = \left(\frac{\psi s \pm t}{a}\right)$. The double sign being at will in this formula, we may suppose it equal to that which occurs in the congruence $\pm t \equiv \chi s \pmod{a}$. If we replace $\pm t$ by χs , one obtains:

$$\left(\frac{t}{a}\right) = \left(\frac{\psi s + \chi s}{a}\right) = \left(\frac{\psi + \chi}{a}\right) \left(\frac{s}{a}\right).$$

On the other hand, one obtains from the congruence $\pm t \equiv \chi s \pmod{a}$, recalling that a is of the form $4n+1$: $\left(\frac{t}{a}\right) = \left(\frac{\chi}{a}\right) \left(\frac{s}{a}\right)$. If one now multiplies member by member this equation and the preceding one, one has $\left(\frac{t^2}{a}\right) = \left(\frac{\chi}{a}\right) \left(\frac{\chi+\psi}{a}\right) \left(\frac{s^2}{a}\right)$ or which comes to the same thing: $\left(\frac{\chi(\chi+\psi)}{a}\right) = 1$, a result which holds when a is a biquadratic residue with respect to p . In the same manner one would find that, if a is not a biquadratic residue with respect to p , one has $\left(\frac{\chi(\chi+\psi)}{a}\right) = -1$.

These results concern the case where φ is not divisible by a . It would remain to treat the case where φ is divisible by a . The analysis to be applied

to this second case, being entirely similar to that which we have set out in detail in what precedes, we shall dispense with developing it here, and shall confine ourselves to stating the conclusion to which it leads:

“If φ is divisible by a , a is or is not a biquadratic residue with respect to p , according as a is of the form $8n + 1$ or of this one: $8n + 5$.”

This result and the preceding one constitute Theorem II stated in the last paragraph of the preceding memoir.

It has doubtless been observed that the statements of Theorems I and II are such that only the root ψ of the even square ψ^2 obtained by decomposing p into two squares enters into them. It would be easy to modify these statements so that they contained only the root φ of the odd square. One would achieve this by following a procedure entirely similar to that which we have set out in what precedes.

The results (δ') and (ϑ') , on which we have relied to establish Theorems I and II, themselves contain infinitely many interesting theorems analogous to the first of M. GAUSS's theorems. One deduces, for example, from the statement (ϑ') , on supposing $a = 5$ in it:

“ p denoting a prime number $20n + 1$, if one puts $p = t^2 + 5u^2$, 5 will or will not be a biquadratic residue with respect to p , according as u is even or odd; on the contrary, if p denotes a prime number $20n + 9$ and one likewise puts $p = t^2 + 5u^2$, 5 will or will not be a biquadratic residue with respect to p , according as u is odd or even.”

The theorems (δ') and (ϑ') concern the equations:

$$t^2 - bu^2 = ps^2, \quad t^2 + au^2 = ps^2.$$

By considerations of the same kind one would find analogous results concerning the equations $t^2 + bu^2 = ps^2$, $t^2 - au^2 = ps^2$.

Treating the first of these equations and then putting, to give an example, $b = 3$, one would obtain this particular theorem:

“ p denoting a prime number $12n + 1$, if one puts $p = t^2 + 3u^2$, t will be an odd number. This being so, I say that 3 will or will not be a biquadratic residue with respect to p , according as t is of the form $12n \pm 1$ or of this one: $12n \pm 5$.”
